

Bounds for Analytical Functions of Matrices

Michel Crouzeix

Abstract. If f is an analytic function bounded on a convex domain of the complex plane and A a square matrix whose spectrum is included in this domain, the function $f(A)$ is well defined. In this paper we study bounds for $\|f(A)\|$ uniform with respect to the functions f bounded by 1, and uniform with respect to the matrices A whose numerical ranges are included in the domain. We show that these bounds are attained and give explicit formulae in some 2-dimensional cases.

Mathematics Subject Classification (2000). 47A12.

Keywords. Numerical range.

1. Introduction

In this paper we work in an Hilbertian context. We use the notations (for generic elements $x, y \in \mathbb{C}^d$, square matrix $A \in \mathbb{C}^{d,d}$):

$\langle x, y \rangle = y^* x = \sum_{j=1}^d x_j \overline{y_j}$ is the usual inner product in $\mathbb{C}^d$,
 $\|x\| = \langle x, x \rangle^{1/2}$ is the Euclidian norm in $\mathbb{C}^d$,
 $\|A\| = \max\{\|Ax\|; x \in \mathbb{C}^d, \|x\| \leq 1\}$ is the operator norm.

Let Ω be an open (non void) convex subset of the complex plane and A be a square matrix with spectrum $Sp(A) \subset \Omega$. We first consider the bound

$$\psi_{\Omega}(A) := \sup\{\|f(A)\|; f \in \mathcal{H}^{\infty}(\Omega), \|f\|_{L^{\infty}(\Omega)} \leq 1\}, \quad (1.1)$$

where $\mathcal{H}^{\infty}(\Omega)$ denotes the Hardy space

$$\mathcal{H}^{\infty}(\Omega) := \{f; f \text{ holomorphic and bounded in } \Omega\}.$$

Let $W(A)$ be the numerical range of A

$$W(A) = \{\langle Ax, x \rangle; x \in \mathbb{C}^d, \|x\| = 1\};$$

we also consider the bound

$$\psi(A) := \sup\{\|p(A)\|; p \text{ polynomial}, \|p\|_{L^{\infty}(W(A))} \leq 1\}. \quad (1.2)$$

These first two bounds are related to a family of analytic functions but to only one matrix. Afterwards we consider a family of matrices

$$\mathcal{M}_d(\Omega) = \{A \in \mathbb{C}^{d,d}; W(A) \subset \overline{\Omega} \text{ and } Sp(A) \subset \Omega\},$$

and the bound

$$\begin{aligned} C_\Omega(d) &:= \sup\{\|f(A)\|; A \in \mathcal{M}_d(\Omega), f \in \mathcal{H}^\infty(\Omega), \|f\|_{L^\infty(\Omega)} \leq 1\}, \\ &= \sup\{\psi_\Omega(A); A \in \mathcal{M}_d(\Omega)\}. \end{aligned} \quad (1.3)$$

It is clear that $C_\Omega(d)$ only depends on d and on the shape of Ω (more precisely $C_\Omega(d) = C_{s(\Omega)}(d)$ if s is a similarity: $s(z) = a + bz$, $a, b \in \mathbb{C}$, $b \neq 0$, or an anti-similarity: $s(z) = a + b\bar{z}$). A theorem due to J. von Neumann [11] asserts that $C_\Omega(d) = 1$ when Ω is a half-plane. In the reference [5] it is proved that $C_\Omega(d)$ is uniformly bounded in d when Ω is convex and bounded, $C_\Omega(d) \leq 3$ if Ω is a disk (see also [3]), and in [4] it is proved that $C_\Omega(d) \leq 2 + 2/\sqrt{3}$ if Ω is a strip or a convex sector of the complex plane.

The organization of the paper is the following. In Section 2 we prove that the bound $\psi_\Omega(A)$ is attained by a function which is a Blaschke product, we give some properties of this bound and an explicit formula in the case of 2×2 matrices. In this last case the numerical range is an ellipse and we express $\psi(A)$ as a function of the eccentricity. In particular we show that $\psi(A) \leq 2$, which implies the following result

Theorem 1.1. *Let A be a 2×2 matrix with entries in $\mathbb{C}$. Then for all polynomials p we have*

$$\|p(A)\| \leq 2 \max_{z \in W(A)} |p(z)|.$$

In Section 3 we show that the bound $C_\Omega(d)$ is attained by a (upper triangular) matrix $A \in \mathcal{M}_d(\Omega)$ and a Blaschke product f in the three following cases: Ω is bounded and its boundary is analytic, Ω is a convex sector or Ω is a strip. We prove that the strip case constant is the limit of the sector constant as the aperture tends to 0 and we include the result that $C_\Omega(d) = 2$ if Ω is a disk (this result has been pointed out to me by Catalin Badea [2], answering a question posed in a previous version of this paper). Section 4 is devoted to the 2×2 case. Since $\psi(A) \leq 2$ we have $C_\Omega(2) \leq 2$; we show that $C_\Omega(2) = 2$ implies that Ω is a disk. Finally we give (quasi)-explicit formulae for $C_\Omega(2)$ when Ω is a sector or a strip of the complex plane.

Conjecture. I have not been able to answer to the following questions :

- does $C_\Omega(d)$ depend on d (for $d \geq 2$) ?
- in the two dimensional case we have proved that the disk corresponds to the worst constant; does this result remain true in any dimension ?

I conjecture a positive answer to the last question. That would imply that Theorem 1 generalizes to any dimension, which would be a very nice result. I end this section with a last minor question :

- in the case “ Ω is an ellipse of eccentricity ε ”, is the constant $C_\Omega(2)$ equal to the right-hand side of equation (2.7) ?

2. Analytic functions of one matrix

Let Ω be a convex subset of the complex plane; we assume $\Omega \neq \emptyset$, $\Omega \neq \mathbb{C}$. Since Ω is simply connected there exists a holomorphic bijection a from Ω onto the open unit disk $D = \{z \in \mathbb{C} ; |z| < 1\}$. From the Osgood-Carathéodory theorem a admits an extension which is a homeomorphism from $\overline{\Omega}$ onto $\overline{D}$. It is convenient to introduce the Blaschke functions

$$b_\zeta(z) = \frac{a(\zeta) - a(z)}{1 - \overline{a(\zeta)} a(z)}, \quad \zeta \in \Omega. \quad (2.1)$$

It is clear that $b_\zeta \in \mathcal{H}^\infty(\Omega) \cap C^0(\overline{\Omega})$, $\|b_\zeta\|_{L^\infty(\Omega)} = 1$, and if $z \in \partial\Omega$ then $|b_\zeta(z)| = 1$. We introduce also the set of finite Blaschke products

$$\mathcal{B}_k(\Omega) = \{f ; f(z) = e^{i\varphi} \prod_{j=1}^r b_{\zeta_j}(z), \quad \varphi \in \mathbb{R}, \zeta_j \in \Omega, j = 1, \dots, r, 0 \leq r \leq k\}.$$

(We use the convention $f(z) = e^{i\varphi}$ if $r = 0$, so $\mathcal{B}_0(\Omega)$ corresponds to constant functions of modulus 1). It can be seen that this space is independent of the choice of the isomorphism a between Ω and D . The following theorem is a consequence of the Nevanlinna-Pick theory [8], [9]

Theorem 2.1. *Let $A \in \mathbb{C}^{d,d}$ be a square matrix with $Sp(A) \subset \Omega$. Then there exists a function $f \in \mathcal{H}^\infty(\Omega)$ such that $\|f(A)\| = \psi_\Omega(A)$ and $\|f\|_{L^\infty(\Omega)} = 1$. Any such a function belongs to $\mathcal{B}_{d-1}(\Omega)$.*

Proof. Let $\{\lambda_j\}_{j=1,\dots,k}$ the set of the distinct eigenvalues of A and $\{r_j\}$ the corresponding multiplicities. Writting A in Jordan form we can see that $f(A)$ only depends on A and on the values $\{f^{(\ell)}(\lambda_j), 0 \leq \ell < r_j, 1 \leq j \leq k\}$. From the compactness property of $\mathcal{H}^\infty(\Omega)$ we deduce that there exists $f \in \mathcal{H}^\infty(\Omega)$ such that $\|f\|_{L^\infty(\Omega)} = 1$ and $\|f(A)\| = \psi_\Omega(A)$. Then let g be the Nevanlinna-Pick interpolant of f , i.e. g is the function which minimizes $\|g\|_{L^\infty(\Omega)}$ among the solutions of

$$g \in \mathcal{H}^\infty(\Omega), \quad g^{(\ell)}(\lambda_j) = f^{(\ell)}(\lambda_j), \quad 0 \leq \ell < r_j, 1 \leq j \leq k. \quad (2.2)$$

It is known [10] [12] that there exists $c \in \mathbb{R}^*$ such that $g/c \in \mathcal{B}_{d-1}(\Omega)$; we have $c = \|g\|_{L^\infty(\Omega)} \leq \|f\|_{L^\infty(\Omega)} = 1$ and g is unique. But (2.2) implies $g(A) = f(A)$, therefore $\psi_\Omega(A) = \|g(A)\| \leq \psi_\Omega(A)/\|g\|_{L^\infty(\Omega)}$. We deduce $\|g\|_{L^\infty(\Omega)} = 1$ and $g = f$. $\square$

Remarks 2.1. We remark on some properties of the function $\psi_\Omega(A)$.

a) If a denotes a holomorphic isomorphism from Ω onto the unit disk D we have $\psi_\Omega(A) = \psi_D(a(A))$. Indeed $f(A) = g(a(A))$ with $g := f \circ a^{-1}$, and clearly f and g have the same maximum norm.

b) If $A = H^{-1}TH$ we have $f(A) = H^{-1}f(T)H$, thus $\psi_\Omega(A) = \psi_\Omega(T)$ as soon as the matrix H is unitary. That allows us to restrict our study to the case of upper triangular matrices without loss of generality.

Lemma 2.2. *The bound $\psi_\Omega(A)$ depends continuously on A and is decreasing with respect to Ω (strictly decreasing if $\psi_\Omega(A) \neq 1$). Furthermore*

$$\psi(A) = \sup\{\psi_\Omega(A) ; W(A) \subset \Omega\}. \quad (2.3)$$

Proof. a) The continuity with respect to A follows from the Cauchy integral representation

$$f(A) - f(A') = \frac{1}{2\pi i} \int_{\mathcal{C}} f(z)((z - A)^{-1} - (z - A')^{-1}) dz,$$

where $\mathcal{C} \subset \Omega$ is an oriented curve surrounding the spectrum of A (and therefore of A' for A' close enough to A), we deduce that $f(A')$ tends to $f(A)$ as $A' \rightarrow A$, uniformly with respect to the functions f bounded by 1.

b) If we have $Sp(A) \subset \Omega \subsetneq \Omega'$, then we have $\psi_\Omega(A) > \psi_{\Omega'}(A)$. Indeed let $f \in \mathcal{B}_{d-1}(\Omega')$ such that $\|f(A)\| = \psi_{\Omega'}(A)$ and $\|f\|_{L^\infty(\Omega')} = 1$. We have clearly $\|f(A)\| \leq \psi_\Omega(A)$ since $\|f\|_{L^\infty(\Omega)} \leq 1$, and we cannot have $\|f(A)\| = \psi_\Omega(A)$ except if $f \in \mathcal{B}_{d-1}(\Omega) \cap \mathcal{B}_{d-1}(\Omega')$, i.e. if f is constant.

Note that this induces a continuity with respect to Ω . Indeed we can assume that $0 \in \Omega$ without loss of generality. Then we set, for $x > 0$, $\Omega_x = x\Omega$, and we clearly have $\psi_{\Omega_x}(A) = \psi_\Omega(x^{-1}A)$. From part a) $\psi_{\Omega_x}(A)$ continuously depends on x . If Ω' is another convex set close to Ω we have

$$|\psi_\Omega(A) - \psi_{\Omega'}(A)| \leq \psi_{\Omega_x}(A) - \psi_{\Omega_y}(A), \quad \text{if } \Omega_x \subset \Omega' \subset \Omega_y \quad \text{and } x < 1 < y.$$

That implies continuity with respect to Ω for the Hausdorff distance.

c) Turning now to the proof of (2.3), we first look at the case when the matrix A is normal. Then we have $\psi(A) = \psi_\Omega(A) = 1$, $\forall \Omega \supset Sp(A)$ and the result is straightforward. In the other cases the interior of $W(A)$ is not empty and we set $\Omega = \text{int}(W(A))$. If $Sp(A) \subset \Omega$ then we have clearly $\psi(A) \leq \psi_\Omega(A)$ and from the previous theorem there exists a function $f \in \mathcal{B}_{d-1}(\Omega)$ such that $\psi_\Omega(A) = \|f(A)\|$. But $\mathcal{B}_{d-1}(\Omega) \subset C^0(\overline{\Omega})$, thus we can find a sequence of polynomials p_n which uniformly tends to f in Ω which proves that $\psi_\Omega(A) \leq \psi(A)$. When $Sp(A) \cap \partial\Omega \neq \emptyset$ we can find [6] a unitary matrix U such that $A = U^* \begin{pmatrix} A_1 & 0 \\ 0 & A_2 \end{pmatrix} U$ with A_2 diagonal and $Sp(A_1) \subset \Omega$. Then it is clear that $\psi(A) = \psi(A_1)$ and for all $\Omega' \supset Sp(A_1)$, $\psi_{\Omega'}(A) = \psi_{\Omega'}(A_1)$. Therefore the result follows from the previous case. $\square$

Now we give an explicit formula in the 2×2 case. In order to express it we introduce the following function

$$\Theta(x, y) := \frac{x + \sqrt{1+x^2}}{y + \sqrt{1+y^2}}.$$

Theorem 2.3. If $T = \begin{pmatrix} \lambda_1 & \gamma \\ 0 & \lambda_2 \end{pmatrix}$, then we have

$$\begin{aligned} \psi_D(T) &= \max\left(1, \Theta\left(\frac{|\gamma|}{|\lambda_1 - \lambda_2|}, \frac{\sqrt{(1-|\lambda_1|^2)(1-|\lambda_2|^2)}}{|\lambda_1 - \lambda_2|}\right)\right), & \text{if } \lambda_1 \neq \lambda_2, \\ \psi_D(T) &= \max\left(1, \frac{|\gamma|}{1-|\lambda_1|^2}\right), & \text{if } \lambda_1 = \lambda_2. \end{aligned}$$

Proof. a) By continuity it is sufficient to consider the case $\lambda_1 \neq \lambda_2$. We define

$$\Psi_1(T) = \frac{|\gamma|}{|\lambda_1 - \lambda_2|}, \quad \Psi_2(T) = \frac{\sqrt{(1-|\lambda_1|^2)(1-|\lambda_2|^2)}}{|\lambda_1 - \lambda_2|},$$

and we recall that

$$a(T) = \begin{pmatrix} a(\lambda_1) & \gamma a[\lambda_1, \lambda_2] \\ 0 & a(\lambda_2) \end{pmatrix}, \quad \text{where } a[\lambda_1, \lambda_2] := \frac{a(\lambda_1) - a(\lambda_2)}{\lambda_1 - \lambda_2}. \quad (2.4)$$

It is easily verified that if a is an automorphism of D the three quantities $\psi_D(T)$, $\Psi_1(T)$, $\Psi_2(T)$ remain invariant if we replace the matrix T by $a(T)$. Indeed we have just to verify this for automorphisms of the form $a(z) = e^{i\varphi}z$, $\varphi \in \mathbb{R}$ and $a(z) = (x - z)/(1 + xz)$, $x \in [0, 1)$, since such mappings a generate the automorphism group of D . Note also that $\psi_D(T)$ does not change if we replace γ by $|\gamma|$ since ψ_D is invariant by a unitary similarity. For all λ_1 and $\lambda_2 \in D$ it is possible to find an automorphism a such that $a(\lambda_1) + a(\lambda_2) = 0$ and $a(\lambda_1) \in (0, 1)$. Therefore it is sufficient to prove the theorem in the case where the matrix T is of the form

$$T = \begin{pmatrix} \lambda & 2\delta \\ 0 & -\lambda \end{pmatrix}, \quad \lambda \in (0, 1), \quad \delta \geq 0. \quad (2.5)$$

b) We now consider this case (2.5). A simple computation shows that $\|T\| = \delta + \sqrt{\lambda^2 + \delta^2}$ and in this situation the statement of the theorem reads

$$\psi_D(T) = \max(1, \|T\|).$$

If $\|T\| \leq 1$, a well-known von Neumann inequality asserts that $\psi_D(T) = 1$, thus we only have to consider the case $\|T\| > 1$. It is clear that $\psi_D(T) \geq \|T\|$ (take $f(z) = z$ in the definition of $\psi_D(T)$). For the converse inequality we set

$$\beta = \frac{1 + \lambda^2}{2\sqrt{\lambda^2 + \delta^2}}, \quad \mu = \frac{1 - \lambda^2 - 2\beta\delta}{2\lambda}, \quad H = \begin{pmatrix} 1 & \mu \\ 0 & \beta \end{pmatrix};$$

then we have

$$B := H^{-1}TH = \begin{pmatrix} \lambda & 2(\lambda\mu + \beta\delta) \\ 0 & -\lambda \end{pmatrix} = \begin{pmatrix} \lambda & 1 - \lambda^2 \\ 0 & -\lambda \end{pmatrix}.$$

This matrix satisfies $\|B\| = 1$, thus $\psi_D(B) = 1$, and consequently $\psi_D(T) \leq \|H\|\psi_D(B)\|H^{-1}\| = \|H\|\|H^{-1}\|$. But this quantity $\|H\|\|H^{-1}\|$ is the largest root of the equation

$$X^2 - \frac{1 + \mu^2 + \beta^2}{\beta}X + 1 = 0. \quad (2.6)$$

If we define $X_1 = \sqrt{\delta^2 + \lambda^2} + \delta$, $X_2 = 1/X_1$, then we have $X_1 X_2 = 1$ and by a simple computation

$$\beta(X_1 + X_2) - \mu^2 - \beta^2 = 1.$$

We deduce that X_1 and X_2 are the two roots of (2.6), thus $\|H\| \|H^{-1}\| = X_1 = \|T\|$, which implies the theorem. $\square$

Corollary 2.4. *If $T = \begin{pmatrix} \lambda_1 & \gamma \\ 0 & \lambda_2 \end{pmatrix}$, and if a denotes a holomorphic bijection from Ω onto the unit disk D , then we have*

$$\begin{aligned} \psi_\Omega(T) &= \max\left(1, \Theta\left(\frac{|\gamma|}{|\lambda_1 - \lambda_2|}, \frac{\sqrt{(1-|a(\lambda_1)|^2)(1-|a(\lambda_2)|^2)}}{|a(\lambda_1) - a(\lambda_2)|}\right)\right), & \text{if } \lambda_1 \neq \lambda_2, \\ \psi_\Omega(T) &= \max\left(1, \frac{|\gamma| |a'(\lambda_1)|}{1 - |a(\lambda_1)|^2}\right), & \text{if } \lambda_1 = \lambda_2. \end{aligned}$$

Proof. From Remark 2.1 a) we have $\psi_\Omega(T) = \psi_D(a(T))$. We apply the previous theorem and use Formula (2.4). $\square$

We turn to the value of the bound $\psi(A)$: we know that $\psi(A) = 1$ if the matrix A is normal. Otherwise it is known that $W(A)$ is an ellipse, and we have

Theorem 2.5. *Let A be a non normal 2×2 matrix with entries in $\mathbb{C}$ and let ε be the eccentricity of the ellipse $W(A)$. Then $\psi(A) = 2$ if $W(A)$ is a disk (i.e. if $\varepsilon = 0$), otherwise*

$$\psi(A) = 2 \exp\left(-\sum_{n \geq 1} \frac{(-1)^{n+1}}{n} \frac{2}{1 + \rho^{4n}}\right), \quad \text{where } \rho = \frac{1 + \sqrt{1 - \varepsilon^2}}{\varepsilon}. \quad (2.7)$$

Proof. We look only at the elliptic case; the disk case will follow by continuity. In order to prove (2.7) we first remark that ψ_A does not change if we replace A by $\alpha A + \beta$, $\alpha \neq 0$, $\alpha, \beta \in \mathbb{C}$, or by a unitarily similar matrix. Therefore we can assume that $A = \begin{pmatrix} 1 & \gamma \\ 0 & -1 \end{pmatrix}$ and furthermore that $\gamma > 0$. Then $W(A)$ is the ellipse of foci $1, -1$ and minor axis γ (see [6] for instance). If ε denotes its excentricity, we have $\gamma = \rho - 1/\rho$ and the major axis is $\rho + 1/\rho$; furthermore (see [7]) the function

$$a(z) = \frac{2z}{\rho} \exp\left(-\sum_{n \geq 1} \frac{(-1)^{n+1}}{n} \frac{2 t_{2n}(z)}{1 + \rho^{4n}}\right),$$

(where t_n denotes the n^{th} Chebyshev polynomial) is the Riemann conforming function mapping $W(A)$ onto $\overline{D}$. We deduce from Corollary 2.4 that, with $\Omega = \text{int}(W(A))$,

$$\psi(A) = \psi_\Omega(A) = \max\left(1, \Theta\left(\frac{\gamma}{2}, \frac{1 - |a(1)|^2}{2|a(1)|}\right)\right) = \rho a(1),$$

which gives (2.7). $\square$

Theorem 1 is a direct consequence of this result since from (9) we clearly have $\psi(A) \leq 2$.

3. Analytic functions of a family of matrices

This section is concerned with the study of the bound $C_\Omega(d)$. For the two dimensional case it follows from the last theorem that

$$C_\Omega(2) = \sup\{\psi_\Omega(A); A \in \mathcal{M}_2(\Omega)\} \leq \sup\{\psi(A); A \in \mathcal{M}_2(\Omega)\} \leq 2. \quad (3.1)$$

We do not know if the same result holds in higher dimensions.

Remark 3.1. In [5] the authors give the majoration $C_D(d) \leq 3$ for the disk and in the general case

$$C_\Omega(d) \leq 3 + \left(\frac{2\pi\delta^2}{|\Omega|}\right)^3,$$

where δ is the diameter of the bounded convex set Ω and $|\Omega|$ its area. This estimate (which is very likely pessimistic) implies the interesting property

for all $f \in \mathcal{H}^\infty(\Omega) \cap C^0(\overline{\Omega})$ the map $A \mapsto f(A)$ is uniformly continuous from $\mathcal{M}_d(\Omega)$ into $\mathbb{C}^{d,d}$;

indeed this property is true when f is a polynomial and the polynomials are dense in $\mathcal{H}^\infty(\Omega) \cap C^0(\overline{\Omega})$.

The following theorems concern the realization of the bound $C_\Omega(\Omega)$.

Theorem 3.1. *Let Ω be a bounded convex subset of the complex plane with an analytic boundary. Then there exists a matrix $A \in \mathcal{M}_d(\Omega)$ and a function $f \in \mathcal{B}_{d-1}(\Omega)$ such that*

$$C_\Omega(d) = \|f(A)\|.$$

Proof. The case $d = 1$ is trivial, $C_\Omega(1) = 1$ and this constant is realized with the constant function $f = 1$. We clearly have $C_\Omega(d) \geq C_\Omega(d-1)$. The case $C_\Omega(d) = C_\Omega(d-1)$ can be treated by induction. Therefore we only look from now on at the case

$$C_\Omega(d) > C_\Omega(d-1). \quad (3.2)$$

From Theorem 2.1 there exists a sequence of $f_n \in \mathcal{B}_{d-1}(\Omega)$ and $A_n \in \mathbb{C}^{d,d}$ satisfying

$$W(A_n) \subset \overline{\Omega}, \quad Sp(A_n) \subset \Omega, \quad \|f_n\|_{L^\infty(\Omega)} = 1, \quad \text{and} \quad \lim_{n \rightarrow \infty} \|f_n(A_n)\| = C_\Omega(d).$$

We write $f_n(z) = \prod_{j=1}^{r_n} b_{\zeta_j^n}(z)$. The sequence of $\{\zeta_j^n \in \Omega\}$ is clearly bounded. The sequence $\{A_n\}$ is also bounded since $\|A_n\| \leq 2 \sup\{|z|; z \in W(A_n)\}$. Therefore (after extraction of a subsequence) we can assume that $A_n \rightarrow A$, $r_n = r$, $\zeta_j^n \rightarrow \zeta_j$. We set $f(z) = \prod_{j=1}^r b_{\zeta_j}(z)$; we have $f \in \mathcal{B}_{d-1}(\Omega)$. Indeed $\zeta_j \in \overline{\Omega}$, and if $\zeta_j \in \partial\Omega$ then the conforming map satisfies $|a(\zeta_j)| = 1$, and thus from (2.1) $b_{\zeta_j}(z) = a(\zeta_j)$ is constant in Ω . We also have $Sp(A) \subset \overline{\Omega}$ and $W(A) \subset \overline{\Omega}$.

We first consider the case where $Sp(A) \subset \Omega$, i.e. $A \in \mathcal{M}_d(\Omega)$. From Remark 3.1 we have $f(A_n) \rightarrow f(A)$ and therefore

$$\begin{aligned} \|f_n(A_n) - f(A)\| &\leq \|f(A_n) - f(A)\| + \|f_n(A_n) - f(A_n)\| \\ &\leq \|f(A_n) - f(A)\| + C_\Omega(d) \|f_n - f\|_{L^\infty(\Omega)} \rightarrow 0, \end{aligned}$$

and consequently $\|f(A)\| = C_\Omega(d)$. Thus the theorem holds if we prove that the opposite case $Sp(A) \cap \partial\Omega \neq \emptyset$ is impossible.

If $Sp(A) \cap \partial\Omega \neq \emptyset$, let $\lambda \in \partial\Omega$ an eigenvalue of A and k its multiplicity. We can assume that the matrices A_n (and thus A) were chosen upper triangular and also that $a_{11} = \dots = a_{kk} = \lambda$. Then we remark that $\forall i = 1, \dots, k, \forall j > i$, we have $a_{ij} = 0$. Indeed (using the canonical basis $\{e_j\}$ in $\mathbb{C}^d$) the condition $W(T) \subset \overline{\Omega}$ implies

$$\forall \mu \in \mathbb{C}, \quad \lambda + \mu a_{ij} + |\mu|^2 a_{jj} = (e_i + \mu e_j)^* A (e_i + \mu e_j) \in (1 + |\mu|^2) \overline{\Omega},$$

which is only compatible with $\lambda \in \partial\Omega$ in the case $a_{ij} = 0$. Now we write the matrices in block form

$$A_n = \begin{pmatrix} T_n & C_n \\ 0 & S_n \end{pmatrix}, \quad A = \begin{pmatrix} \lambda I_k & 0 \\ 0 & S \end{pmatrix}.$$

For n large enough $Sp(T_n) \cap Sp(S_n) = \emptyset$, therefore we can define U_n as the unique solution of

$$U_n S_n - T_n U_n = C_n,$$

which we can also write as

$$U_n = (C_n + (T_n - \lambda I_k) U_n) (S_n - \lambda I_{d-k})^{-1}.$$

Since $C_n \rightarrow 0$ and $T_n - \lambda I_k \rightarrow 0$ we deduce $U_n \rightarrow 0$. We remark now that

$$A_n = \begin{pmatrix} I & -U_n \\ 0 & I \end{pmatrix} \begin{pmatrix} T_n & 0 \\ 0 & S_n \end{pmatrix} \begin{pmatrix} I & U_n \\ 0 & I \end{pmatrix},$$

thus

$$f_n(A_n) = \begin{pmatrix} I & -U_n \\ 0 & I \end{pmatrix} \begin{pmatrix} f_n(T_n) & 0 \\ 0 & f_n(S_n) \end{pmatrix} \begin{pmatrix} I & U_n \\ 0 & I \end{pmatrix}.$$

But we have $\|f_n(T_n)\| \leq C_\Omega(k)$ and $\|f_n(S_n)\| \leq C_\Omega(d-k)$. If $k < d$ this implies $\|f_n(A_n)\| \leq (1 + \|U_n\|) C_\Omega(d-1) (1 + \|U_n\|)$. Taking the limit as $n \rightarrow \infty$ we deduce $C'_\Omega(d) \leq C_\Omega(d-1)$ which is in contradiction with (3.2). Therefore the only possible case is $k = d$ i.e. $A = \lambda I_d$.

It remains to show that this case $A = \lambda I_d$ is impossible. Note that $a \in C^0(\overline{\Omega})$. As in Remark 3.1 we deduce from $A_n \rightarrow \lambda I_d$ that $a(A_n) \rightarrow a(\lambda) I_d$, therefore if $\zeta_j \neq \lambda$ it follows from $\zeta_j^n \rightarrow \zeta_j$ that

$$b_{\zeta_j^n}(A_n) = (a(\zeta_j^n) - a(A_n))(1 - a(\overline{\zeta_j^n}) a(A_n))^{-1} \rightarrow b_{\zeta_j}(\lambda) I_d.$$

If $\zeta_j = \lambda$, we use the smoothness assumption on the boundary. The function a admits a holomorphic extension into a neighborhood of λ . Let C be a circle of

center λ included in this neighborhood. We deduce from the uniform convergence of $b_{\zeta_j^n}$ towards $b_\lambda = a(\lambda) = b_\lambda(\lambda)$ on C and from the Cauchy representation

$$b_{\zeta_j^n}(A_n) = \frac{1}{2\pi i} \int_C (z - A_n)^{-1} b_{\zeta_j^n}(z) dz,$$

that $b_{\zeta_j^n}(A_n) \rightarrow b_\lambda(\lambda) I_d$. Therefore we have $\lim_{n \rightarrow \infty} f_n(A_n) = f(\lambda) I_d$, and consequently $C_\Omega(d) = 1$ which is also in contradiction with (3.2) since $C_\Omega(d-1) \geq 1$. $\square$

Theorem 3.2. *Let Ω be a strip or a convex sector of the complex plane. Then there exists a matrix $A \in \mathcal{M}_d(\Omega)$ and a function $f \in \mathcal{B}_{d-1}(\Omega)$ with $\|f\|_{L^\infty(\Omega)} = 1$ such that*

$$C_\Omega(d) = \|f(A)\|.$$

Proof. We first consider the strip case. Without loss of generality we can assume that Ω is the strip

$$\Omega = B = \{z \in \mathbb{C}; |\operatorname{Im} z| \leq 1\}, \quad \text{and we use } a(z) = \tanh \frac{\pi}{4} z.$$

Note that (after a slight modification which is the multiplication by a unitary constant) the Blaschke functions are now $b_\zeta(z) = \frac{\operatorname{sh}(\frac{\pi}{4}(\zeta-z))}{\operatorname{ch}(\frac{\pi}{4}(\bar{\zeta}-z))}$. As in the bounded case we can restrict our study to the case

$$C_\Omega(d) > C_\Omega(d-1).$$

Then we consider a sequence of $f_n \in \mathcal{B}_{d-1}(B)$ and $A_n \in \mathbb{C}^{d,d}$ satisfying

$$W(A_n) \subset \overline{B}, \quad Sp(A_n) \subset B, \quad \|f_n\|_{L^\infty(\Omega)} = 1, \quad \text{and} \quad \lim_{n \rightarrow \infty} \|f_n(A_n)\| = C_\Omega(d). \quad (3.3)$$

We write $f_n(z) = \prod_{j=1}^{r_n} b_{\zeta_j^n}(z)$. We can assume A_n upper triangular, with the eigenvalues $\lambda_j^n = a_{jj}^n$ ordered such that $\operatorname{Re} \lambda_j^n$ increase with j . We can also suppose $\operatorname{trace}(A_n) = 0$ (indeed replacing (A_n, f_n) by $(A_n - x_n, f_n(\cdot + x_n))$ for $x_n \in \mathbb{R}$ does not modify (3.3)).

We first prove that the sequence A_n is bounded. Otherwise (perhaps after extraction of a subsequence) there will exist an index k such that $\operatorname{Re}(\lambda_{k+1}^n - \lambda_k^n) \rightarrow \infty$. Then we write in block form

$$A_n = \begin{pmatrix} T_n & C_n \\ 0 & S_n \end{pmatrix}, \quad \text{with } T_n \in \mathbb{C}^{k,k}.$$

Since $Sp(S_n) \cap Sp(T_n) = \emptyset$ we can define U_n as the unique solution of

$$U_n S_n - T_n U_n = C_n,$$

which we can also write ($\bullet$ denotes the componentwise product)

$$U_n = H_n \bullet (C_n + (T_n - \operatorname{diag}(T_n)) U_n - U_n (S_n - \operatorname{diag}(S_n))),$$

where $H_n = (h_{ij}^n) \in \mathbb{C}^{k,d-k}$ is defined by $h_{ij}^n = \frac{1}{\lambda_i^n - \lambda_{j+k}^n}$.

We have $W(A_n) \subset \overline{B}$, i.e. $\|A_n - A_n^*\| \leq 2$. Since A_n is triangular, this implies that $A_n - \text{diag}(A_n)$ is bounded. Therefore C_n , $T_n - \text{diag}(T_n)$ and $S_n - \text{diag}(S_n)$ are bounded. Together with $H_n \rightarrow 0$, this implies $U_n \rightarrow 0$. We remark now that

$$A_n = \begin{pmatrix} I & -U_n \\ 0 & I \end{pmatrix} \begin{pmatrix} T_n & 0 \\ 0 & S_n \end{pmatrix} \begin{pmatrix} I & U_n \\ 0 & I \end{pmatrix},$$

thus

$$f_n(A_n) = \begin{pmatrix} I & -U_n \\ 0 & I \end{pmatrix} \begin{pmatrix} f_n(T_n) & 0 \\ 0 & f_n(S_n) \end{pmatrix} \begin{pmatrix} I & U_n \\ 0 & I \end{pmatrix}.$$

This implies $\|f_n(A_n)\| \leq (1 + \|U_n\|)^2 C_\Omega(d-1)$, and by taking the limit at ∞ gives $C'_\Omega(d) \leq C_\Omega(d-1)$ which is in contradiction with (3.2). Therefore the sequence A_n is bounded.

We turn now to the sequences $\{\zeta_j^n\}$. After extraction of a subsequence we can assume that $r_n = r$ and that for each j one of these three possibilities occurs : “the sequence $\{\zeta_j^n\}$ is bounded”, “ $\lim \text{Re } \zeta_j^n = +\infty$ ”, “ $\lim \text{Re } \zeta_j^n = -\infty$ ”. We can also assume that $\limsup \|b_{\zeta_j^n}(A_n)\| > 1$ (otherwise we could replace f_n by $f_n/b_{\zeta_j^n}$ and (3.3) would be still satisfied). We consider now the cases “ $\lim \text{Re } \zeta_j^n = \varepsilon \infty$ ” ($\varepsilon = \pm 1$). Then we have $b_{\zeta_j^n} \rightarrow \varepsilon$ uniformly on each compact subset of $\mathbb{C}$. Consequently $\lim \|b_{\zeta_j^n}(A_n)\| = 1$, possibility which has been excluded. Therefore we have proved that the family $\{\zeta_j^n\}$ is bounded. The remaining part of the proof is the same as for Theorem 3.1.

We turn now to the proof of Theorem 3.2 in the case of a convex sector Ω . We can assume that $\Omega = S_\alpha := \{z \in \mathbb{C}; z \neq 0 \text{ and } |\arg z| < \alpha\}$, for some $0 < \alpha \leq \frac{\pi}{2}$, and that $C'_\Omega(d) > C_\Omega(d-1)$. We consider a sequence of functions $f_n \in \mathcal{B}_{d-1}(S_\alpha)$ and upper triangular matrices $A_n \in \mathbb{C}^{d,d}$ satisfying

$$W(A_n) \subset \overline{S}_\alpha, \quad Sp(A_n) \subset S_\alpha, \quad \|f_n\|_{L^\infty(S_\alpha)} = 1, \quad \lim_{n \rightarrow \infty} \|f_n(A_n)\| = C_{S_\alpha}(d). \quad (3.4)$$

We write $f_n(z) = \prod_{j=1}^{r_n} b_{\zeta_j^n}(z)$. We also assume that $|\det(A_n)| = 1$ (this is possible since replacing (A_n, f_n) by $(x_n A_n, f_n(\cdot/x_n))$ with $x_n > 0$ does not affect (3.4)).

We introduce now $L_n = \frac{1}{\alpha} \log(A_n)$, $g_n(\zeta) = f_n(e^{\alpha \zeta})$. Then we have $g_n \in \mathcal{B}_{d-1}(S_B)$, $L_n \in \mathbb{C}^{d,d}$ satisfying

$$W(L_n) \subset \overline{B}, \quad Sp(L_n) \subset B, \quad \|g_n\|_{L^\infty(B)} = 1, \quad \text{and} \quad \lim_{n \rightarrow \infty} \|g_n(L_n)\| = C_{S_\alpha}(d). \quad (3.5)$$

(the proof of $W(L_n) \subset \overline{B}$ will be given in Theorem 3.3). The argument used in the strip case allows us to show that the sequence $\{L_n\}$ is bounded; consequently the sequences $\{A_n\}$ and $\{(A_n)^{-1}\}$ are bounded. Similarly we obtain the boundedness of the sequences $\{\zeta_j^n\}$ and $\{(\zeta_j^n)^{-1}\}$. We conclude as previously. $\square$

Theorem 3.3. *Let S_α be a sector with angle $0 \leq 2\alpha \leq \pi$ and B be a strip of the complex plane. Then the constant $C_{S_\alpha}(d)$ is a decreasing function of α . We have $C_{S_\alpha}(d) \leq C_B(d)$ and $C_B(d) = \lim_{\alpha \rightarrow 0} C_{S_\alpha}(d)$.*

Proof. a) We first prove that $W(A) \subset S_\alpha$ implies $W(A^s) \subset S_{s\alpha}$ for all $0 < s < 1$. For this we introduce the two half planes $\Pi_{\alpha+}$ and $\Pi_{\alpha-}$

$$\Pi_{\alpha+} = \{z \in \mathbb{C}; \operatorname{Re} e^{i(\pi/2-\alpha)} z > 0\}, \quad \Pi_{\alpha-} = \{z \in \mathbb{C}; \operatorname{Re} e^{i(-\pi/2+\alpha)} z > 0\},$$

so that $S_\alpha = \Pi_{\alpha+} \cap \Pi_{\alpha-}$. It is clear that the function $a(z) = z^s$ (we choose the branch such that $a(1) = 1$) maps the half plane $\Pi_{\alpha\pm}$ into $\Pi_{s\alpha\pm}$. Since $W(A) \subset \Pi_{\alpha\pm}$ we have (still a von Neumann result) $W(a(A)) \subset a(\Pi_{\alpha\pm})$, i.e. $W(A^s) \subset S_{s\alpha}$.

b) A first consequence is that $C_{S_\alpha}(d)$ decreases with α . Indeed we choose a $d \times d$ matrix A such that $W(A) \subset \overline{S}_\alpha$ and $\psi_{S_\alpha}(A) = C_{S_\alpha}(d)$. We have, for $0 < s < 1$, $\psi_{S_\alpha}(A) = \psi_{S_{s\alpha}}(A^s) \leq C_{S_{s\alpha}}(d)$ since $W(A^s) \subset \overline{S}_{s\alpha}$.

c) With the same choice of matrix we set $L = \frac{1}{\alpha} \log(A)$. It is clear that $a(z) = \frac{1}{\alpha} \log z$ is an isomorphism from S_α onto B . Thus we have $\psi_{S_\alpha}(A) = \psi_B(L)$. We will get $\psi_B(L) \leq C_B(d)$ i.e. $C_{S_\alpha}(d) \leq C_B(d)$ if we prove that $W(L) \subset \overline{B}$.

For this we remark that $L = \lim_{n \rightarrow \infty} \frac{n}{\alpha} (A^{1/n} - I)$. Therefore if $u \in \mathbb{C}^d$ satisfies $\|u\| = 1$ we have (since $W(A^{1/n}) \subset S_{\alpha/n}$)

$$|\operatorname{Im} \langle Lu, u \rangle| = \lim_{n \rightarrow \infty} \frac{n}{\alpha} |\operatorname{Im} \langle A^{1/n} u, u \rangle| \leq \lim_{n \rightarrow \infty} \frac{n}{\alpha} \tan\left(\frac{\alpha}{n}\right) \operatorname{Re} \langle A^{1/n} u, u \rangle = 1,$$

which gives $W(L) \subset \overline{B}$.

d) We know that $C_B(d) \geq \lim_{\alpha \rightarrow 0} C_{S_\alpha}(d)$, and it remains to prove that $C_B(d) \leq \lim_{\alpha \rightarrow 0} C_{S_\alpha}(d)$. For this it is sufficient to show that for any $d \times d$ matrix L with $W(L) \subset B$ it is possible to choose $\alpha > 0$ sufficiently small such that $\psi_B(L) \leq C_{S_\alpha}(d)$. Let $L = M + iN$ be such a matrix with M and N self-adjoint; since $W(L) \subset B$ we have $\|N\| < 1$. We set $A = \exp(\alpha L)$; then $\psi_B(L) = \psi_{S_\alpha}(A)$. Furthermore if $u \in \mathbb{C}^d$ satisfies $\|u\| = 1$ we have

$$\langle Au, u \rangle = 1 + \alpha \langle Mu, u \rangle + i\alpha \langle Nu, u \rangle + O(\alpha^2),$$

thus

$$\tan(\arg(\langle Au, u \rangle)) = \alpha \langle Nu, u \rangle + O(\alpha^2),$$

and consequently $|\tan(\arg(\langle Au, u \rangle))| \leq \alpha (\|N\| + O(\alpha))$ (the term $O(\alpha)$ tends to 0 uniformly in u). Therefore we can choose α such that $\alpha (\|N\| + O(\alpha)) \leq \tan \alpha$, and then we have $W(A) \subset \overline{S}_\alpha$ and consequently $\psi_B(L) = \psi_{S_\alpha}(A) \leq C_{S_\alpha}(d)$. $\square$

Theorem 3.4. *In the disk case, for $d \geq 2$ we have $C_D(d) = 2$.*

Proof. I owe this elegant proof to [2]. We start from a result of [1] which asserts that, if $A \in \mathcal{M}_d(D)$, then there exist a hermitian matrix B and a unitary matrix U such that $A = 2 \cos(B)U \sin(B)$. We consider the continuous function $g(x) = \max(1, 2|\cos x|)$ and define (using the continuous functional calculus for self-adjoint operators) the matrices $H = g(B)$ and $T = H^{-1}AH$. We clearly have

$$\|H\| \leq 2, \quad \|H^{-1}\| \leq 1, \quad \|\sin(B)H\| \leq 1, \quad 2\|H^{-1}\cos(B)\| \leq 1, \quad \text{thus } \|T\| \leq 1.$$

If $f \in \mathcal{H}^\infty(D)$ we have $f(A) = Hf(T)H^{-1}$, therefore

$$\|f(A)\| \leq \|H\| \|f(T)\| \|H^{-1}\| \leq 2 \|f\|_{L^\infty(D)}.$$

With the choice $f(z) = z$ and $A = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}$ we see that the bound 2 is attained. $\square$

4. The two dimensional case

In this section we consider 2×2 matrices.

Theorem 4.1. *For any open convex Ω ($\neq \emptyset$ and $\neq \mathbb{C}$) of the complex plane we have $C_\Omega(2) \leq 2$. Furthermore the equality $C_\Omega(2) = 2$ implies that Ω is a disk.*

Proof. The inequality has been seen previously (3.1). If there exists a matrix $A \in \mathcal{M}_2(\Omega)$ such that $C_\Omega(2) = \psi_\Omega(A)$ (which is the case if the boundary of Ω is analytic, cf. Theorem 3.1) then

$$C_\Omega(2) = \psi_\Omega(A) \leq \psi(A) \leq 2.$$

But the first inequality is strict if $\overline{\Omega} \neq W(A)$, and from (2.7) the second is strict if Ω is not a disk. Therefore in the analytic case $C_\Omega(2) = 2$ implies that Ω is a disk.

We turn now to the case when $C_\Omega(2) = 2$ and the constant $C_\Omega(2)$ is not attained. Arguing as in the proof of Theorem 3.1 we can find a sequence of upper triangular matrices $A_n \in \mathcal{M}_2(\Omega)$ and of functions f_n with the following properties

$$f_n(z) = \frac{a(\zeta_n) - a(z)}{1 - a(\zeta_n)a(z)}, \quad \|f_n(A_n)\| \rightarrow 2, \quad \zeta_n \rightarrow \zeta, \quad \text{and } A_n \rightarrow \zeta Id.$$

We set

$$A_n = \begin{pmatrix} \lambda_1^n & 2\varepsilon_n \\ 0 & \lambda_2^n \end{pmatrix}, \quad \lambda_n = \frac{\lambda_1^n + \lambda_2^n}{2}, \quad B_n = \begin{pmatrix} \lambda_n & 2\varepsilon_n \\ 0 & \lambda_n \end{pmatrix}.$$

We have $\psi_{A_n} \rightarrow 2$ and we deduce from (2.7) that the eccentricity of $W(A_n)$ tends to 0, thus $\frac{\lambda_1^n - \lambda_2^n}{\varepsilon_n} \rightarrow 0$. Using the Cauchy representation

$$f_n(A_n) - f_n(B_n) = \frac{1}{2\pi i} \int_{\partial W(B_n)} f_n(z) ((z - A_n)^{-1} - (z - B_n)^{-1}) dz,$$

we get $f_n(A_n) - f_n(B_n) \rightarrow 0$. Since the disk $W(B_n) = \{z; |z - \lambda_n| \leq \varepsilon_n\}$ satisfies $W(B_n) \subset W(A_n) \subset \overline{\Omega}$ and since Ω is convex it is possible (at least for n large enough) to find a point $\mu_n \in \partial W(B_n)$ such that the disk $\Delta_n = \{z; |z - \mu_n| < \varepsilon_n\}$ satisfies $\Delta_n \subset \Omega$. We consider now the set $W = \{z; z = a + \xi, 0 < a < 1, |\xi| < 1\}$ which is the convex hull of the union of the unit disk D and the disk $\Delta = 1 + D$, and we define

$$g_n(z) := f_n(\lambda_n + z(\mu_n - \lambda_n)), \quad B := \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}.$$

We clearly have $\|g_n(B)\| = \|f_n(B_n)\| \rightarrow 2$ and $\|g_n\|_{L^\infty(W)} \leq 1$. We can extract a subsequence such that $g_n \rightarrow g \in \mathcal{H}^\infty(W)$ (in C^∞ of any compact subset of W). Then we have $\|g(B)\| = 2$ and $\|g\|_{L^\infty(W)} \leq 1$. From $\psi_B = 2$ and Theorem 2.1 we deduce $g \in \mathcal{B}_1(D)$, i.e. we can write $g(z) = e^{i\varphi} (z - \mu)(1 - \overline{\mu}z)^{-1}$ with $\mu \in D$.

But then $2 = \|g(B)\| = 1 + \sqrt{1+|\mu|^2}$ implies $\mu = 0$, i.e. $g(z) = e^{i\varphi} z$, which is in contradiction with $\|g\|_{L^\infty(W)} \leq 1$. $\square$

We now give explicit formulae for the sector and for the strip in 2 dimensions.

Theorem 4.2. *In the 2-dimensional case we have the following formulae*

$$\begin{aligned} C_{S_\alpha}(2) &= \sup_{y>0} \tau_\alpha(y), & \text{if } 0 < \alpha < \frac{\pi}{4}, \\ C_{S_\alpha}(2) &= \sqrt{2}, & \text{if } \alpha = \frac{\pi}{4}, \\ C_{S_\alpha}(2) &= \sup_{c \in (0,1)} \sigma_\alpha(c), & \text{if } \frac{\pi}{4} < \alpha \leq \frac{\pi}{2}, \end{aligned}$$

where

$$\begin{aligned} \tau_\alpha(y) &:= \tanh \frac{\pi}{4} y \frac{\sin \alpha + \sqrt{\sin^2 \alpha + \operatorname{sh}^2 \alpha y}}{\operatorname{sh} \alpha y}, \\ \text{and} \\ \sigma_\alpha(c) &:= \tan \frac{\pi}{4} c \frac{\sin \alpha + \sqrt{\sin^2 \alpha - \sin^2 \alpha c}}{\sin \alpha c}. \end{aligned}$$

Proof. In this proof we will use the generic upper triangular matrix T and the isomorphism a from S_α onto D defined by

$$T = \begin{pmatrix} \lambda_1 & \gamma \\ 0 & \lambda_2 \end{pmatrix}, \quad \text{and} \quad a = \frac{z^{\pi/2\alpha} - \lambda_2^{\pi/2\alpha}}{z^{\pi/2\alpha} - \overline{\lambda_2}^{\pi/2\alpha}}.$$

By an argument of continuity it is sufficient to only look at the case $\lambda_1 \neq \lambda_2$. Since $a(\lambda_2) = 0$, it follows from Corollary 2.4 that

$$\psi_{S_\alpha}(T) = \max \left(1, \Theta \left(\frac{|\gamma|}{|\lambda_1 - \lambda_2|}, \frac{\sqrt{1 - |a(\lambda_1)|^2}}{|a(\lambda_1)|} \right) \right). \quad (4.1)$$

Note that $W(T)$ is an ellipse (cf. [6] for instance) with foci λ_1 and λ_2 and minor axis $|\gamma|$. We remark that it is sufficient to consider the matrices T such that $W(T)$ is tangent to the two edges of S_α , indeed otherwise we could find another sector $S = \zeta + S_\alpha$ such that $W(T) \subset S \subsetneq S_\alpha$ but then, from Lemma 2.2, $\psi_{S_\alpha}(T) < \psi_S(T) = \psi_{S_\alpha}(T - \zeta)$. Therefore

$$C_{S_\alpha}(2) = \max \{ \psi_{S_\alpha}(T) ; T \in \mathcal{M}_d, W(T) \text{ bitangent to } \partial S_\alpha \}. \quad (4.2)$$

From the second Poncelet theorem the two edges of S_α (tangent to $W(T)$) and the two focus lines $O\lambda_1, O\lambda_2$ have the same bisectrix, therefore $\arg \lambda_1 + \arg \lambda_2 = 0$. Thus we can set

$$\lambda_1 = \rho e^{\alpha(y+i\theta)}, \quad \lambda_2 = \rho e^{-\alpha(y+i\theta)}, \quad \text{with } y \in \mathbb{R}, \quad |\theta| < 1.$$

The edge $\{r e^{i\alpha} ; r > 0\}$ being tangent to $W(T)$ we have

$$\begin{aligned} 0 &= \max_{\|u\|=1} \operatorname{Im} e^{-i\alpha} \langle Tu, u \rangle \\ &= \max_{\|u\|=1} (-\rho e^{\alpha y} |u_1|^2 \sin \alpha (1-\theta) - \rho e^{-\alpha y} |u_2|^2 \sin \alpha (1+\theta) + \operatorname{Im} \gamma e^{-i\alpha} \bar{u}_1 u_2), \\ &= |\gamma|/2 - \rho \sqrt{\sin \alpha (1-\theta) \sin \alpha (1+\theta)}. \end{aligned}$$

Thus $|\gamma| = 2\rho\sqrt{\cos^2\alpha\theta - \cos^2\alpha}$, furthermore $|\lambda_1 - \lambda_2| = 2\rho|\operatorname{sh}\alpha(y + i\theta)| = 2\rho\sqrt{\operatorname{ch}^2\alpha y - \cos^2\alpha\theta}$, this gives

$$\frac{|\gamma|}{|\lambda_1 - \lambda_2|} = \frac{\sqrt{\cos^2\alpha\theta - \cos^2\alpha}}{\sqrt{\operatorname{ch}^2\alpha y - \cos^2\alpha\theta}}.$$

Similarly we get

$$\frac{\sqrt{1 - |a(\lambda_1)|^2}}{|a(\lambda_1)|} = \frac{2\sqrt{\operatorname{Re}\lambda_1^{\pi/2\alpha} \operatorname{Re}\lambda_2^{\pi/2\alpha}}}{|\lambda_1^{\pi/2\alpha} - \lambda_2^{\pi/2\alpha}|} = \frac{\cos\frac{\pi\theta}{2}}{\sqrt{\operatorname{ch}^2\frac{\pi y}{2} - \cos^2\frac{\pi\theta}{2}}}.$$

Putting this into (4.1), we deduce if $\psi_{S_\alpha}(T) > 1$

$$\begin{aligned} \psi_{S_\alpha}(T) &= \frac{\sqrt{\operatorname{ch}^2\alpha y - \cos^2\alpha} + \sqrt{\cos^2\alpha\theta - \cos^2\alpha}}{\sqrt{\operatorname{ch}^2\alpha y - \cos^2\alpha\theta}} \frac{\sqrt{\operatorname{ch}^2\frac{\pi y}{2} - \cos^2\frac{\pi\theta}{2}}}{\operatorname{ch}\frac{\pi y}{2} + \cos\frac{\pi\theta}{2}}, \\ &= \left(\frac{\sqrt{\operatorname{ch}^2\alpha y - \cos^2\alpha} + \sqrt{\cos^2\alpha\theta - \cos^2\alpha}}{\sqrt{\operatorname{ch}^2\alpha y - \cos^2\alpha} - \sqrt{\cos^2\alpha\theta - \cos^2\alpha}} \right)^{1/2} \left(\frac{\operatorname{ch}\frac{\pi y}{2} - \cos\frac{\pi\theta}{2}}{\operatorname{ch}\frac{\pi y}{2} + \cos\frac{\pi\theta}{2}} \right)^{1/2}, \end{aligned}$$

and coming back to (4.2) we get

$$C_{S_\alpha}(2) = \max\{(\phi_\alpha(y, \theta))^{1/2}; y \geq 0, \theta \in [0, 1]\}, \quad \text{with } \phi_\alpha(y, \theta) := (\psi_{S_\alpha}(T))^2. \quad (4.3)$$

More explicitly

$$\begin{aligned} \phi_\alpha(0, 0) &= \frac{\pi \sin \alpha}{2\alpha}, \\ \phi_\alpha(y, \theta) &= \frac{\sqrt{\operatorname{ch}^2\alpha y - \cos^2\alpha} + \sqrt{\cos^2\alpha\theta - \cos^2\alpha}}{\sqrt{\operatorname{ch}^2\alpha y - \cos^2\alpha} - \sqrt{\cos^2\alpha\theta - \cos^2\alpha}} \frac{\operatorname{ch}\frac{\pi y}{2} - \cos\frac{\pi\theta}{2}}{\operatorname{ch}\frac{\pi y}{2} + \cos\frac{\pi\theta}{2}}. \end{aligned}$$

In the case $\alpha = \frac{\pi}{4}$ simple calculations give

$$\phi_{\pi/4}(y, \theta) := \frac{(\sqrt{\operatorname{ch}\frac{\pi}{2}y} + \sqrt{\cos\frac{\pi}{2}\theta})^2}{\operatorname{ch}\frac{\pi}{2}y + \cos\frac{\pi}{2}\theta},$$

and using the Cauchy-Schwarz inequality we deduce $C_{S_{\pi/4}}(2) = \sqrt{2} = \sqrt{\phi_{\pi/4}(0, 0)}$.

We turn to the other values of α . We remark that the function ϕ_α is C^∞ except maybe in the point $(y, \theta) = (0, 0)$. We know from Theorem 3.2 that the constant $C_{S_\alpha}(2)$ is realized for a matrix T . Therefore there is a point (y, θ) such that either $(y, \theta) = (0, 0)$, or $\nabla\phi_\alpha(y, \theta) = 0$. This last relation may be written

$$\begin{aligned} \frac{\frac{\alpha \operatorname{sh} \alpha y}{\sqrt{\operatorname{ch}^2 \alpha y - \cos^2 \alpha}} \frac{\operatorname{ch} \alpha y}{\operatorname{ch}^2 \alpha y - \cos^2 \alpha}}{\frac{\alpha \sin \alpha \theta}{\sqrt{\cos^2 \alpha \theta - \cos^2 \alpha}} \frac{\cos \alpha \theta}{\operatorname{ch}^2 \alpha y - \cos^2 \alpha}} \frac{2\sqrt{\cos^2 \alpha \theta - \cos^2 \alpha}}{\operatorname{ch}^2 \alpha y - \cos^2 \alpha} &= \frac{\pi}{2} \operatorname{sh} \frac{\pi y}{2} \frac{2 \cos \frac{\pi \theta}{2}}{\operatorname{ch}^2 \frac{\pi y}{2} - \cos^2 \frac{\pi \theta}{2}}, \\ \frac{\alpha \sin \alpha \theta}{\sqrt{\cos^2 \alpha \theta - \cos^2 \alpha}} \frac{\cos \alpha \theta}{\operatorname{ch}^2 \alpha y - \cos^2 \alpha} \frac{2\sqrt{\operatorname{ch}^2 \alpha y - \cos^2 \alpha}}{\operatorname{ch}^2 \alpha y - \cos^2 \alpha} &= \frac{\pi}{2} \sin \frac{\pi \theta}{2} \frac{2 \operatorname{ch} \frac{\pi y}{2}}{\operatorname{ch}^2 \frac{\pi y}{2} - \cos^2 \frac{\pi \theta}{2}}. \end{aligned}$$

If $y \neq 0$ and $\theta \neq 0$ this implies

$$\frac{\operatorname{sh} 2\alpha y}{\sin 2\alpha \theta} \frac{\cos^2 \alpha \theta - \cos^2 \alpha}{\operatorname{ch}^2 \alpha y - \cos^2 \alpha} = \frac{\tanh \frac{\pi y}{2}}{\tan \frac{\pi \theta}{2}},$$

or equivalently $\mu_\alpha(y) = \nu_\alpha(\theta)$ with

$$\mu_\alpha(y) := \tanh \frac{\pi y}{2} \frac{\operatorname{ch} 2\alpha y - \cos 2\alpha}{\operatorname{sh} 2\alpha y}, \quad \nu_\alpha(\theta) = \tan \frac{\pi \theta}{2} \frac{\cos 2\alpha \theta - \cos 2\alpha}{\sin 2\alpha \theta}.$$

But this is impossible since for $y > 0$ and $\theta \in (0, 1)$ we have (see Lemma 4.3 below)

$$\nu_\alpha(\theta) < \frac{\pi \sin^2 \alpha}{2\alpha} < \mu_\alpha(y), \quad \text{if } 0 < \alpha < \frac{\pi}{4},$$

$$\nu_\alpha(\theta) > \frac{\pi \sin^2 \alpha}{2\alpha} > \mu_\alpha(y), \quad \text{if } \frac{\pi}{4} < \alpha < \frac{\pi}{2}.$$

Therefore we have necessarily $y = 0$ or $\theta = 0$. Note that $\phi_\alpha(y, 0) = (\tau_\alpha(y))^2$ and $\phi_\alpha(0, \theta) = (\sigma_\alpha(\theta))^2$; therefore going back to (4.3) we get

$$C_{s_\alpha}(2) = \max \left(\sup \{ \tau_\alpha(y) ; y > 0 \}, \sup \{ \sigma_\alpha(c) ; c \in (0, 1) \} \right). \quad (4.4)$$

But we have $\tau_\alpha(0) = \sigma_\alpha(0)$ and (see the next lemma) τ_α is decreasing for $\alpha > \frac{\pi}{4}$ and σ_α is decreasing for $\alpha < \frac{\pi}{4}$. We deduce the theorem from (4.4). $\square$

Lemma 4.3. *The functions ν_α , μ_α , τ_α and σ_α have the following properties*

- a) μ_α is increasing for $\alpha < \pi/4$, decreasing for $\alpha > \pi/4$,
- b) ν_α is decreasing for $\alpha < \pi/4$, increasing for $\alpha > \pi/4$,
- c) τ_α is decreasing for $\alpha > \pi/4$,
- d) σ_α is decreasing for $\alpha < \pi/4$.

Proof. a) The derivative μ'_α has the same sign as

$$\begin{aligned} f_\alpha(y) &:= \operatorname{ch}^2 \frac{\pi}{2} y \operatorname{sh}^2 2\alpha y \mu'_\alpha(y) \\ &= \frac{\pi}{4} \operatorname{sh} 4\alpha y - \alpha \operatorname{sh} \pi y - \cos 2\alpha \left(\frac{\pi}{2} \operatorname{sh} 2\alpha y - \alpha \operatorname{sh} \pi y \operatorname{ch} 2\alpha y \right) \end{aligned}$$

We clearly have $f_\alpha(0) = 0$ and

$$f'_\alpha(y) = \pi \alpha \cos 2\alpha g_\alpha(y), \quad \text{where}$$

$$\begin{aligned} g_\alpha(y) &:= \frac{\operatorname{ch} 4\alpha y - \operatorname{ch} \pi y}{\cos 2\alpha} + \operatorname{ch} 2\alpha y (\operatorname{ch} \pi y - 1) + \frac{2\alpha}{\pi} \operatorname{sh} \pi y \operatorname{sh} 2\alpha y, \\ &= -\frac{2 \operatorname{sh}(\frac{\pi}{2} - 2\alpha)y \operatorname{sh}(\frac{\pi}{2} + 2\alpha)y}{\cos 2\alpha} + 2 \operatorname{sh} \frac{\pi}{2} y (\operatorname{sh} \frac{\pi}{2} y \operatorname{ch} 2\alpha y + \frac{2\alpha}{\pi} \operatorname{ch} \frac{\pi}{2} y \operatorname{sh} 2\alpha y) \\ &= \operatorname{sh} \frac{\pi y}{2} \left[\left(1 + \frac{2\alpha}{\pi}\right) \operatorname{sh} \left(\frac{\pi}{2} + 2\alpha\right)y + \left(1 - \frac{2\alpha}{\pi}\right) \operatorname{sh} \left(\frac{\pi}{2} - 2\alpha\right)y - 2 \frac{\operatorname{sh}(\frac{\pi}{2} - 2\alpha)y \operatorname{sh}(\frac{\pi}{2} + 2\alpha)y}{\operatorname{sh} \frac{\pi}{2} y \cos 2\alpha} \right]. \end{aligned}$$

We shall prove that $\forall y > 0$, $\forall \alpha \in (0, \frac{\pi}{2})$, we have $g_\alpha(y) > 0$, thus μ'_α has the same sign as $\cos 2\alpha$ and part a) of the lemma follows. (Note that $g_0(y) = g_{\pi/2}(y) = 0$ and $g_{\pi/4}(y) = \frac{3}{2} \operatorname{sh} \pi y (\operatorname{sh} \frac{\pi}{2} y - \frac{4}{3}y)$).

It is clear that $g_\alpha(y) > 0$ if $1 + \frac{2\alpha}{\pi} - 2 \frac{\operatorname{sh}(\frac{\pi}{2} - 2\alpha)y}{\operatorname{sh} \frac{\pi}{2} y \cos 2\alpha} \geq 0$; note that we have

$$\forall y > 0, \quad \frac{|\operatorname{sh}(\frac{\pi}{2} - 2\alpha)y|}{\operatorname{sh} \frac{\pi}{2} y} \leq \frac{|\pi/2 - 2\alpha|}{\pi/2},$$

therefore we have $g_\alpha(y) \geq 0$ as soon as $1 + \frac{2\alpha}{\pi} - 2 \frac{|\pi/2 - 2\alpha|}{\pi/2 \cos 2\alpha} \geq 0$, i.e. if we have the inequality $|\cos 2\alpha| \geq 2 \frac{|\pi/2 - 2\alpha|}{\pi/2 + \alpha}$. This is the case for $\frac{\pi}{6} \leq \alpha \leq \frac{\pi}{2}$. We turn now

to the case $0 < \alpha < \frac{\pi}{6}$. Then we have

$$\frac{\operatorname{sh}(\frac{\pi}{2}-2\alpha)y \operatorname{sh}(\frac{\pi}{2}+2\alpha)y}{\operatorname{sh}^2 \frac{\pi}{2}y} = \frac{\operatorname{sh}^2 \frac{\pi}{2}y - \operatorname{sh}^2 2\alpha y}{\operatorname{sh}^2 \frac{\pi}{2}y} < 1.$$

Therefore

$$g_\alpha(y) \geq \operatorname{sh} \frac{\pi y}{2} \left[(1 + \frac{2\alpha}{\pi}) \operatorname{sh}(\frac{\pi}{2}+2\alpha)y + (1 - \frac{2\alpha}{\pi}) \operatorname{sh}(\frac{\pi}{2}-2\alpha)y - \frac{2}{\cos 2\alpha} \sqrt{\operatorname{sh}(\frac{\pi}{2}-2\alpha)y \operatorname{sh}(\frac{\pi}{2}+2\alpha)y} \right].$$

We deduce $g_\alpha(y) > 0$ if $(\frac{1}{\cos 2\alpha})^2 \leq (1 + \frac{2\alpha}{\pi})(1 - \frac{2\alpha}{\pi})$, i.e. $\frac{2\alpha^2}{\pi^2} < \sin 4\alpha$ which is the case since $0 < \alpha < \frac{\pi}{6}$. A similar proof gives the part b).

c) The derivative τ'_α has the same sign as

$$\begin{aligned} \operatorname{ch}^2 \frac{\pi y}{4} \operatorname{sh}^2 \alpha y \tau'_\alpha(y) &= \frac{\pi}{4} (\sin \alpha + \sqrt{\sin^2 \alpha + \operatorname{sh}^2 \alpha y}) \operatorname{sh} \alpha y + \\ &\quad + \alpha \operatorname{sh} \frac{\pi y}{4} \operatorname{ch} \frac{\pi y}{4} \left(\frac{\operatorname{sh}^2 \alpha y \operatorname{ch} \alpha y}{\sqrt{\sin^2 \alpha + \operatorname{sh}^2 \alpha y}} - \sin \alpha \operatorname{ch} \alpha y - \operatorname{ch} \alpha y \sqrt{\sin^2 \alpha + \operatorname{sh}^2 \alpha y} \right) \\ &= \frac{\sin \alpha + \sqrt{\sin^2 \alpha + \operatorname{sh}^2 \alpha y}}{\sqrt{\sin^2 \alpha + \operatorname{sh}^2 \alpha y}} \left(\frac{\pi}{4} \operatorname{sh} \alpha y \sqrt{\sin^2 \alpha + \operatorname{sh}^2 \alpha y} - \frac{\alpha \sin \alpha}{2} \operatorname{sh} \frac{\pi y}{2} \operatorname{ch} \alpha y \right). \end{aligned}$$

Therefore τ'_α has the same sign as

$$\begin{aligned} f_\alpha(y) &:= \frac{2 \operatorname{sh}^2 \alpha y (\sin^2 \alpha + \operatorname{sh}^2 \alpha y)}{\alpha^2 \sin^2 \alpha} - \frac{8}{\pi^2} \operatorname{sh}^2 \frac{\pi y}{2} \operatorname{ch}^2 \alpha y \\ &= \frac{\operatorname{ch} 2\alpha y - 1}{\alpha^2} + \frac{\operatorname{ch} 4\alpha y - 4 \operatorname{ch} 2\alpha y + 3}{4\alpha^2 \sin^2 \alpha} - \frac{\operatorname{ch}(\pi+2\alpha)y + \operatorname{ch}(\pi-2\alpha)y + 2 \operatorname{ch} \pi y - 2 \operatorname{ch} 2\alpha y - 2}{\pi^2} \\ &= - \sum_{k \geq 2} \beta_k(\alpha) \frac{y^{2k}}{(2k)!}, \end{aligned}$$

with

$$\beta_k(\alpha) := \frac{(\pi+2\alpha)^{2k} + (\pi-2\alpha)^{2k} + 2\pi^{2k} - 2(2\alpha)^{2k}}{\pi^2} - \frac{(4\alpha)^{2k} - 4(2\alpha)^{2k} \cos^2 \alpha}{4\alpha^2 \sin^2 \alpha}.$$

Note that $\beta_1(\alpha) = 4 - 4 = 0$. The decreasing property of τ_α will follow from the inequalities

$$\beta_k(\alpha) > 0, \quad \text{for all } k \geq 2 \quad \text{and} \quad \beta \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right).$$

To prove them we first remark that $\beta_2(\alpha) = 4(\pi^2 + 8\alpha^2 - \frac{12\alpha^2}{\sin^2 \alpha}) > 0$, for $\alpha \in (\frac{\pi}{4}, \frac{\pi}{2})$. For $k \geq 3$ we have

$$\beta_k(\alpha) \geq \frac{(\pi+2\alpha)^{2k}}{\pi^2} - \frac{(4\alpha)^{2k}}{4\alpha^2 \sin^2 \alpha} \geq (4\alpha)^{2k-6} \left[\frac{(\pi+2\alpha)^6}{\pi^2} - \frac{(4\alpha)^6}{4\alpha^2 \sin^2 \alpha} \right].$$

In order to complete the proof it is sufficient to show that

$$\gamma := (\pi+2\alpha)^6 \sin^2 \alpha - 1024 \pi^2 \alpha^4 > 0.$$

It is easily seen that for $\alpha \in (\frac{\pi}{4}, \frac{\pi}{2})$ we have $\sin^2 \alpha (\pi+2\alpha)^2 \geq 16\alpha^2$, therefore

$$\gamma \geq 16\alpha^2 [(\pi+2\alpha)^4 - 64\pi^2 \alpha^2] = 16\alpha^2 [(\pi+2\alpha)^2 + 8\pi\alpha] (\pi-2\alpha)^2 > 0.$$

A similar proof gives part d). $\square$

By Theorem 3.3 and Theorem 4.2, taking the limit as $\alpha \rightarrow 0$ gives

Corollary 4.4. *In the two-dimensional case we have for the strip domain*

$$C_B(2) = \max_{y>0} \frac{1+\sqrt{1+y^2}}{y} \operatorname{th} \frac{\pi y}{4} \simeq 1.58765981\dots$$

References

- [1] T. Ando, Structure of operators with numerical range 1, *Acta Sci. Math. (Szeged)* **34** (1973) 11–15.
- [2] C. Badea, Personal communication.
- [3] C. A. Berger, J. G. Stampell, Mapping theorems for the numerical range, *Amer. J. Math.* **89** (1967) 1047–1055.
- [4] M. Crouzeix, B. Delyon, Some estimates for analytic functions of band or sectorial operators, *submitted* (2002).
- [5] B. Delyon, F. Delyon, Generalization of Von Neumann's spectral sets and integral representation of operators, *Bull. Soc. Math. Fr.* **127** no. 1 (1999) 25–42.
- [6] K. E. Gustafson, D. K. M. Rao, *Numerical Range*, Universitext, Springer-Verlag, (1997).
- [7] P. Henrici, *Applied and computational complex analysis*, Wiley-Interscience, vol. 3, (1985).
- [8] R. Nevanlinna, Über beschränkte Funktionen, die in gegebenen Punkten vorgeschriebene Werte annehmen, *Ann. Acad. Sci. Fenn. Ser. A, no. 1* **13** (1919).
- [9] G. Pick, Über die Beschränkungen analytischer Funktionen, welche durch vorgegebene Funktionwerte bewirkt werden, *Math. Ann.* **77** (1916) 7–23.
- [10] D. Sarason, Generalized interpolation in H^∞ , *Trans. Amer. Math. Soc.* **127** (1967) 179–203.
- [11] J. von Neumann, Eine Spektraltheorie für allgemeine Operatoren eines unitären Raumes, *Math. Nachrichten* **4** (1951) 258–281.
- [12] J. L. Walsh, Interpolation and Approximation by Rational Functions in the Complex Domain, *Amer. Math. Soc. Colloq. Publ. Providence, RI* **20** (1956).

Michel Crouzeix

Institut de Recherche Mathématique de Rennes, UMR 6625

Université de Rennes 1

Campus de Beaulieu

35042 Rennes Cedex

France

e-mail: `michel.crouzeix@univ-rennes1.fr`

Submitted: February 15, 2002

Revised: July 15, 2002